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Determinants of the capacity of dryland ecosystems to store soil carbon under altered fire regimes

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Abstract

Widespread changes in the intensity and frequency of fires across the globe are altering the terrestrial carbon (C) sink[1–4]. Although the changes in ecosystem C have been reasonably well quantified for plant biomass pools[5–7], an understanding of the determinants of fire-driven changes in soil organic C (SOC) across broad environmental gradients remains unclear, especially in global drylands[3,4,7–9]. Here, we combined multiple datasets and original field sampling of fire manipulation experiments to evaluate where and why fire changes SOC the most, built a statistical model to estimate historical changes in SOC, and compared these estimates to simulations from ecosystem models. We found that drier ecosystems experienced larger relative changes in SOC than humid ecosystems—in some cases exceeding losses from plant biomass pools—primarily explained by high fire-driven declines in tree biomass inputs in dry ecosystems. Ecosystem models provided more mixed insight into potential SOC changes because many models underestimated the SOC changes in drier ecosystems. Upscaling our statistical model predicted that soils in 1.57 million km^2 savanna-grassland regions experiencing declines in burned area over the past ca. two decades may have 23% more SOC, equating to 1.78 PgC in topsoils. Consequently, ongoing declines in fire frequencies have likely created an extensive carbon sink in the soils of global drylands that may have been underestimated by ecosystem models.

Full Text

Fire-driven changes in SOC arising from altered fire frequencies are hypothesized to be predicted by how much fire directly combusts SOC[4,10], and indirectly alters plant biomass inputs to soils and decomposition of residual SOC post-fire[11–17]. In drier and warmer ecosystems, which dominate global burned area, most SOC is in the mineral horizon where heat rapidly dissipates[18] and little direct combustion of SOC occurs[16,19,20]. In these drier sites, fire-driven shifts in plant biomass inputs, especially from trees[21–23], are thought to determine changes in SOC stored in the mineral horizon[24–26]. Consequently, increases in fire frequency may drive large SOC losses in climates with low precipitation and/or seasonal rainfall, where water constrains tree growth and post-fire biomass recovery[6,27–29], relative to ecosystems in climates where biomass recovery is faster. In addition to climate, soil texture and mineralogy can modify post-fire decomposition rates[30–32] and promote higher C losses in ecosystems with coarse textured soils[12]. Thus, we hypothesize that water availability, temperature, and soil texture all act to modify the effect of repeated burning on SOC storage in the mineral horizon.

Global data to evaluate these hypotheses are lacking because there have yet to be studies examining repeated burning effects on SOC and plant biomass in parallel across broad climatic and ecological gradients, despite comparisons within individual ecosystems[4,24,33]. Thus, models used to simulate the effects of fire regime changes on ecosystems, such as fire-enabled Dynamic Global Vegetation Models[34] (DGVMs), lack a clear benchmark for evaluating how well they simulate SOC responses across environmental gradients[7]. Here, we examine the factors that determine the magnitude of SOC losses or gains in the mineral horizon when fire frequencies change, evaluate whether DGVMs capture spatial

patterns in fire effects on SOC storage, and estimate the potential impact of recent regional changes in fire frequencies on SOC storage.

To evaluate the determinants of fire effects on SOC, we conducted a meta-analysis to identify the environmental variables relating to how multi-decadal alterations of fire frequency impact SOC storage in the mineral horizon using data from experiments in 53 sites containing 434 replicate plots. Within these sites, we compared the effect of repeated burning at different frequencies relative to unburned plots or plots burned at lower frequencies over the same period (Table S1, *Supplemental Information, SI*). We focused our analyses on ecosystems that account for the majority of both total burned area and recent changes in fire frequency (savannas, grasslands, as well as seasonal woodlands and forests)[2] (Figure S1).

Globally, our meta-analysis demonstrated that the most important climatic and edaphic variables explaining fire effects on SOC in the mineral horizon were aridity, precipitation seasonality, mean annual temperature, and silt content, with larger relative changes in SOC in drier and cooler environments on coarsely textured soils (r^2=0.82, p<0.001, Table S2; *SI*). Using model selection on a mixed-effects meta-regression model, we identified the important environmental variables, and then fit the most parsimonious model to the data to illustrate the influence of these variables on fire effects (Table S2; *SI*). Variables related to experimental design and overall ecosystem type were also incorporated to better isolate environmental variables (SI). Relative to unburned plots, SOC in burned plots was 17±10% lower in sites with average aridity and 37±23% lower in sites with high aridity (p<0.001, Figure 1a, Table S2; aridity defined as the ratio between precipitation/potential evapotranspiration, *SI*). Annual precipitation seasonality was the second most important environmental variable in the statistical model, with twice as large fire-driven declines in high vs. average seasonality sites (27±8% lower vs. 13±9% lower, respectively, p<0.001, Figure 1b, Table S2). Relative SOC losses were also greater in sites with cooler temperatures and coarser textured soils (p<0.01 and p=0.068, respectively), although these environmental variables were less important than aridity and precipitation seasonality according to variable importance analyses (Table S2). These variables were identified when we controlled for differences arising from broad ecosystem classification and experimental duration and fire frequency[3] (Table S2, Figures S2). Taken together, water availability was the most important environmental factor for explaining the relative change in SOC with altered fire frequencies.

To test the hypothesis that fire-driven changes in SOC could be attributed to changes in tree biomass inputs across sites, we focused on savanna-grasslands and analysed 74 plots across seven sites in our meta-analysis with data on soil ^{13}C. We used ^{13}C as a proxy for tree biomass inputs in these sites because C3 tree biomass has a lower ^{13}C than C4 grass biomass. Thus, SOC ^{13}C is commonly used to quantify tree biomass inputs relative to C4 grass inputs in savannas[24,25,35].

Fire-driven changes in ^{13}C illustrate that larger decreases in SOC in dry environments is linked with lower tree biomass inputs to soils. Comparing ^{13}C values across sites and in unburned vs. burned plots illustrated that frequent burning and higher aridity caused a shift from C3 tree- to C4 grass-derived

biomass inputs. The proportion of SOC from C3 trees was, on average, 55±11% lower in frequently burned relative to unburned plots. The losses of C3-derived inputs positively correlated with the losses in SOC stocks across sites, pointing to changes in woody biomass inputs to soils driving changes in SOC storage but the relative magnitude of change varied across sites (r^2=0.71, p=0.01, Figure 2a, Table S3). Aridity explained the variability in ^{13}C changes across sites because the most arid sites experienced the strongest fire-driven declines in C3-derived inputs (r^2=0.58, p=0.029, Figure 2b), consistent with fire causing the largest relative changes in SOC in drier climates. Thus, the degree to which fire changes plant biomass inputs, specifically trees, into soils helps explains patterns in SOC responses to fire.

To evaluate estimates of fire effects on SOC at the global scale, we analyzed whether an ensemble of seven fire-enabled DGVMs were able to recreate the biogeographical trends in fire effects found in our empirical findings (model details in refs.[7,34] and *SI*, Figure S3 for global maps). We used model experiments that were similar in concept to our field experiments: comparing simulations of SOC in a world with fire vs. world without fire. Rather than compare model-based estimates of SOC fluxes with data at individual sites, we compare the within-model relationships between fire effects and water availability gradients with the predictions from our empirical model across the same gradients. Generally, the DGVMs predicted that areas experiencing the largest differences in burned area between the fire vs. no fire simulations also experienced the largest changes in SOC (Figure S4). However, the models were inconsistent in their simulated patterns of SOC sensitivity to fire across aridity and precipitation seasonality gradients (Figures 3,S5,S6). In savanna-grasslands, which account for 70% of global burned area, only two of the seven models (LPJ-GUESS-BLAZE and CTEM models) correctly recreated the empirically determined relationships between the sensitivity of SOC to fire and both precipitation seasonality and aridity (Figures 3,S5,S6). Although we cannot isolate the role of model-based differences in simulating burned area across climates, these results suggest a DGVM ensemble is likely biased towards underestimating fire-driven changes in SOC in drier regions.

To estimate the potential area over which frequent burning could limit mineral SOC, we scaled up our statistical model of fire effects on SOC to savanna-grasslands. To estimate what areas may be either losing or gaining SOC, we extrapolated observed trends in burned area for ca. two decades[2] to identify areas of increasing or decreasing fire frequency and used the environmental covariates and SOC content derived from other global maps (*SI*) to estimate potential SOC changes. In 2.30 million km^2 where burned area is tending to decline, SOC has potentially risen by 23% (Figure 4a). In 1.38 million km^2 where burned area is tending to rise, SOC has potentially declined by 25% (Figure 4b). By multiplying these relative values with total SOC stocks, we estimate reductions in burning from 1998-2015 resulted in a gain of 1.78 PgC while more frequent fires resulted in a loss of 1.14 PgC, for a net-change of 0.64 PgC, or a flux of 0.038 PgC yr^{-1}.

While previous research has highlighted the capacity of savanna-grasslands soils to serve as a C sink[26], our study identifies multiple mechanisms that explain the wide variability in SOC responses to fire across drylands as a whole. With this information, informed management choices can be made that implement

nature-based climate solutions[12]. We demonstrate the relative SOC potential is roughly double in drier savanna-grasslands, but potentially comes with a cost of high tree encroachment, which can reduce biodiversity—a key consideration that may be offset through management of browsing herbivores, which we did not consider here.

We did not focus on systems with either intense crown wildfires such as spruce boreal forest or smoldering ground fires in peatlands. Although these ecosystems can store large amounts of SOC, they burn relatively rarely (albeit increasing in frequency)[12], and the factors that determine direct combustion of SOC are well characterised[4]. Meanwhile, savannas have lower per-area SOC stocks, but given their large spatial extent and frequent burning, SOC stocks underlying savannas that burn equate to ~14.5 PgC[12], which could be relevant for future terrestrial C storage. Our sink estimate is small relative to the residual land C sink (3.1±0.6 PgC yr^{-1})[36], but given the lack of saturation observed over the duration of experiments in our dataset (~60 years), SOC in drylands may be a persistent sink.

Declarations

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Figures

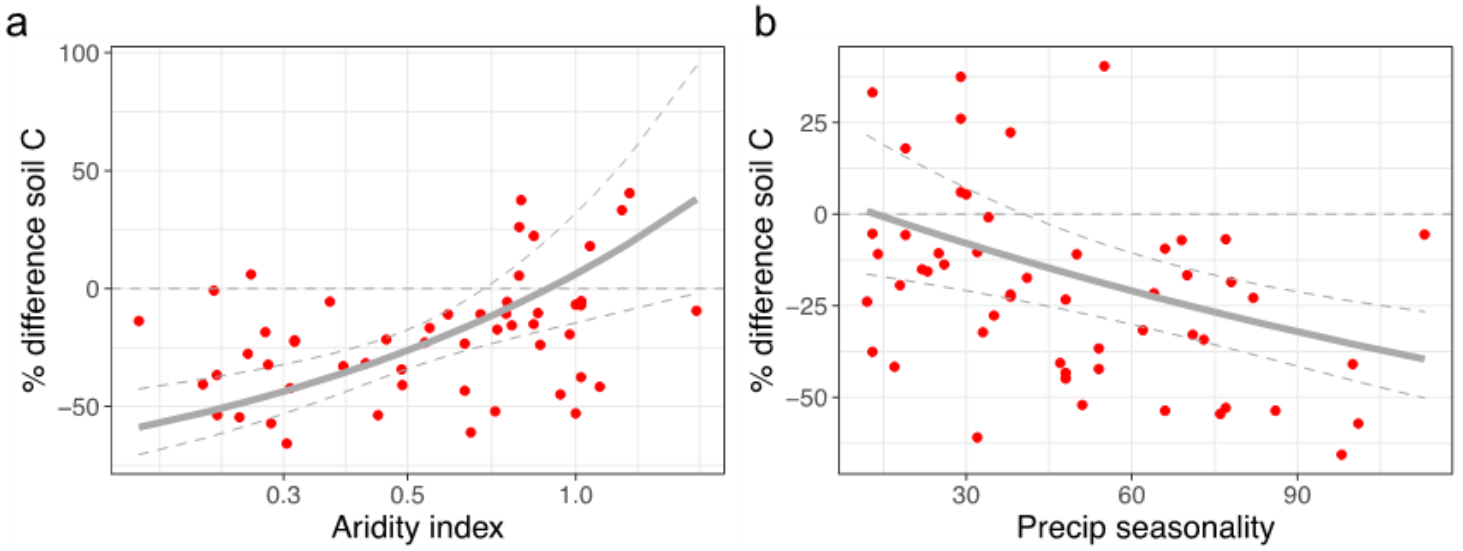


Figure 1

Water availability modifies the effect of fire on soil organic carbon (SOC). Results from meta-regression of the top model illustrating how environmental conditions influenced the percent difference in SOC concentrations in the burned versus unburned plots (lower values thus signify a fire-driven loss). a) Fire effects as a function of aridity (precipitation/potential evapotranspiration), with a low aridity index indicating dry conditions (*SI*). b) Precipitation seasonality is the coefficient of variation of monthly precipitation within a year multiplied by 100. All dashed lines indicate 95% confidence intervals of the model fit. Importance of all variables in the model selection presented in Table S2.

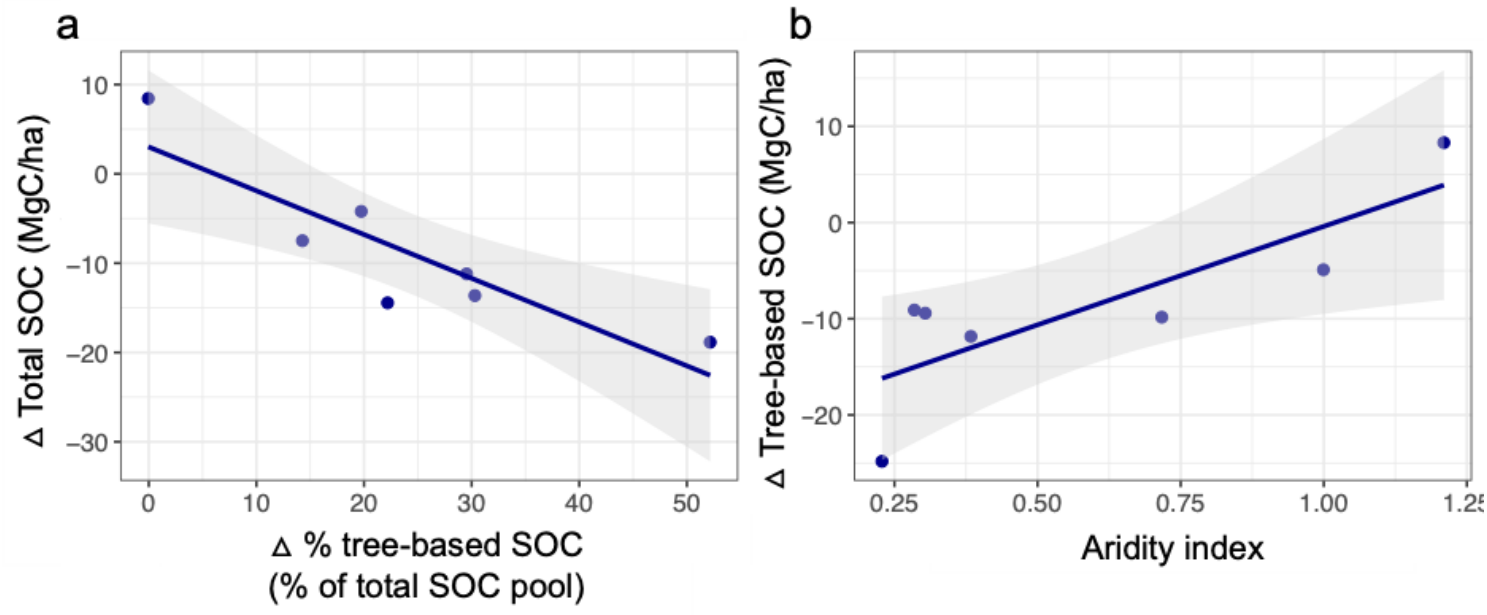


Figure 2

Fire effects on soil organic carbon (SOC) are predicted by changes in tree-based SOC which track the aridity gradient a) Difference in the total SOC between burned – unburned and the percent of SOC derived from tree biomass (stocks in 0-20 cm of the mineral horizon). b) Aridity index (lower values are drier) and the difference in SOC from trees between burned – unburned. Lines are linear regressions with shaded area representing the standard error. These averages are based on n=74 plots distributed across the sites.

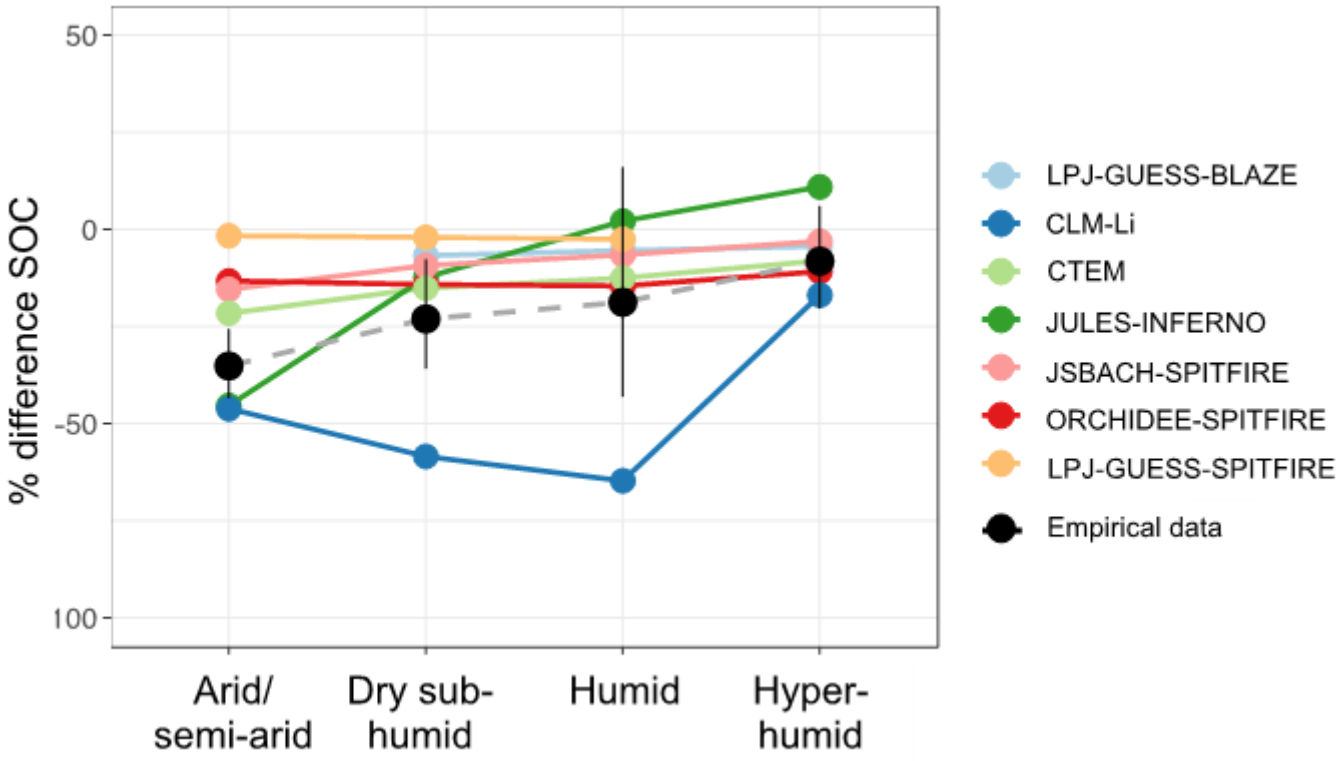


Figure 3

Most models underpredict the stronger relative effects of fire on soil organic carbon (SOC) in drier environments. Comparison between model simulations and empirical data of fire-driven changes in SOC across aridity classes. Coloured lines represent means from different Dynamic Global Vegetation Models (DGVMs) and the black line represents the data with error bar illustrating the 95% confidence intervals. DGVMs calculate the percent difference by comparing the simulations with fire vs. a 'world without fire' (described in SI). Aridity index is calculated as the ratio between precipitation and potential evapotranspiration.

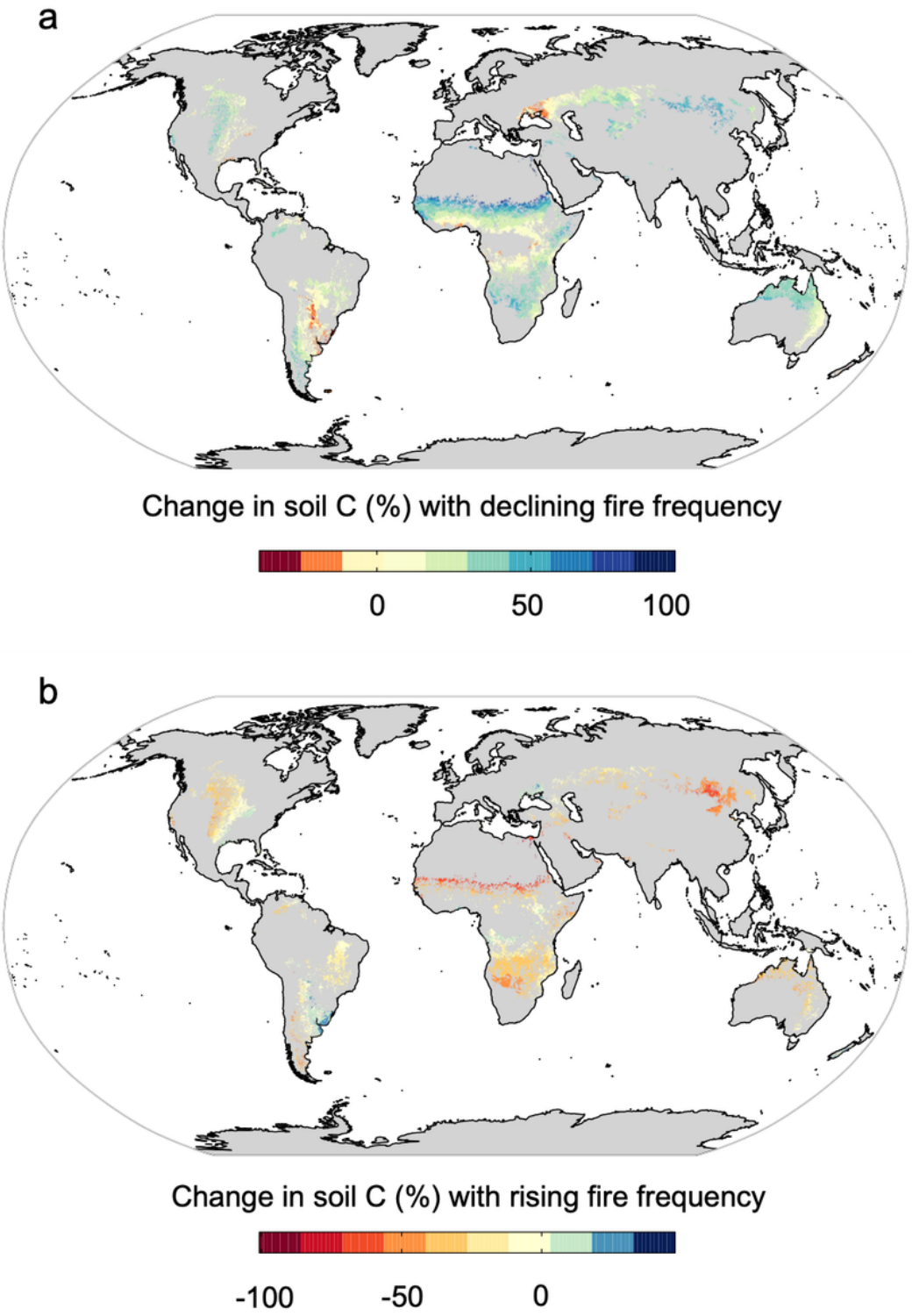


Figure 4

Sensitivity of soil carbon to changes in fire frequencies across savanna-grasslands globally. Upscaling our analysis to savanna-grasslands worldwide across the distribution of the areas where our model predicted a change in soil organic carbon (C) under observed trends in burned area. a) Predicted gain in soil C in areas with declining fire frequency, with a positive % illustrating a gain in soil C under the new fire regime. b) Predicted change in soil C with higher fire frequency, with a lower % illustrating a loss in soil C

in the frequently burned. We estimated changes only in areas within environmental conditions used in our training model.

Supplementary Files

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